



PERSONAL DATA AND PRIVACY IN A NETWORKED WORLD

DSCI 531
Spring 2025

Kristina Lerman



What is private data

Personally Identifiable Information (PII): data that can be used to identify a specific individual, either directly or indirectly.

- Direct identifiers include:
 - Full name
 - Social Security number, Driver's license number, Passport number
 - Email address, Phone numbers
 - Physical address, ZIP code
- Indirect identifiers (which can identify someone when combined with other information) include:
 - Date of birth, Place of birth
 - Gender
 - Race/ethnicity
 - Medical record numbers, Biometric data (fingerprints, facial recognition)

PII is protected by various privacy laws and regulations worldwide, requiring special handling, protection, and often explicit consent for collection and processing.



Private Data and Research

Ethics of human subjects research (HSR) are closely connected to PII, govern use of social data at universities

- HSR is regulated by Institutional Review Board (IRB)
 - IRBs evaluate research protocols specifically for how PII will be: Collected and destroyed (necessity and minimization)
 - Stored (security measures)
 - Used (scope limitations)
 - Shared (data transfer protocols)
- Informed consent and risks
 - Presence of PII increases risk to subjects
 - IRB weighs risks and benefits
 - “Informed consent” protects subject’s autonomy – have to understand potential risks and agree to them
- IRB and social media research
 - Increasing protections for data publicly shared on social media



Private Data and Regulation

Key regulations protecting PII

- US
 - **HIPAA (Health Insurance Portability and Accountability Act)**: Protects medical information and health records
 - **FCRA (Fair Credit Reporting Act)**: Regulates credit information
 - **FERPA (Family Educational Rights and Privacy Act)**: Protects student records
 - **COPPA (Children's Online Privacy Protection Act)**: Protects information of children under 13
- EU
 - **GDPR (General Data Protection Regulation)**: Comprehensive EU-wide privacy regulation
- Other international
 - Each regulation has different definitions of PII, enforcement mechanisms, and requirements for regarding consent, data minimization, security measures, breach notifications, and data subject rights



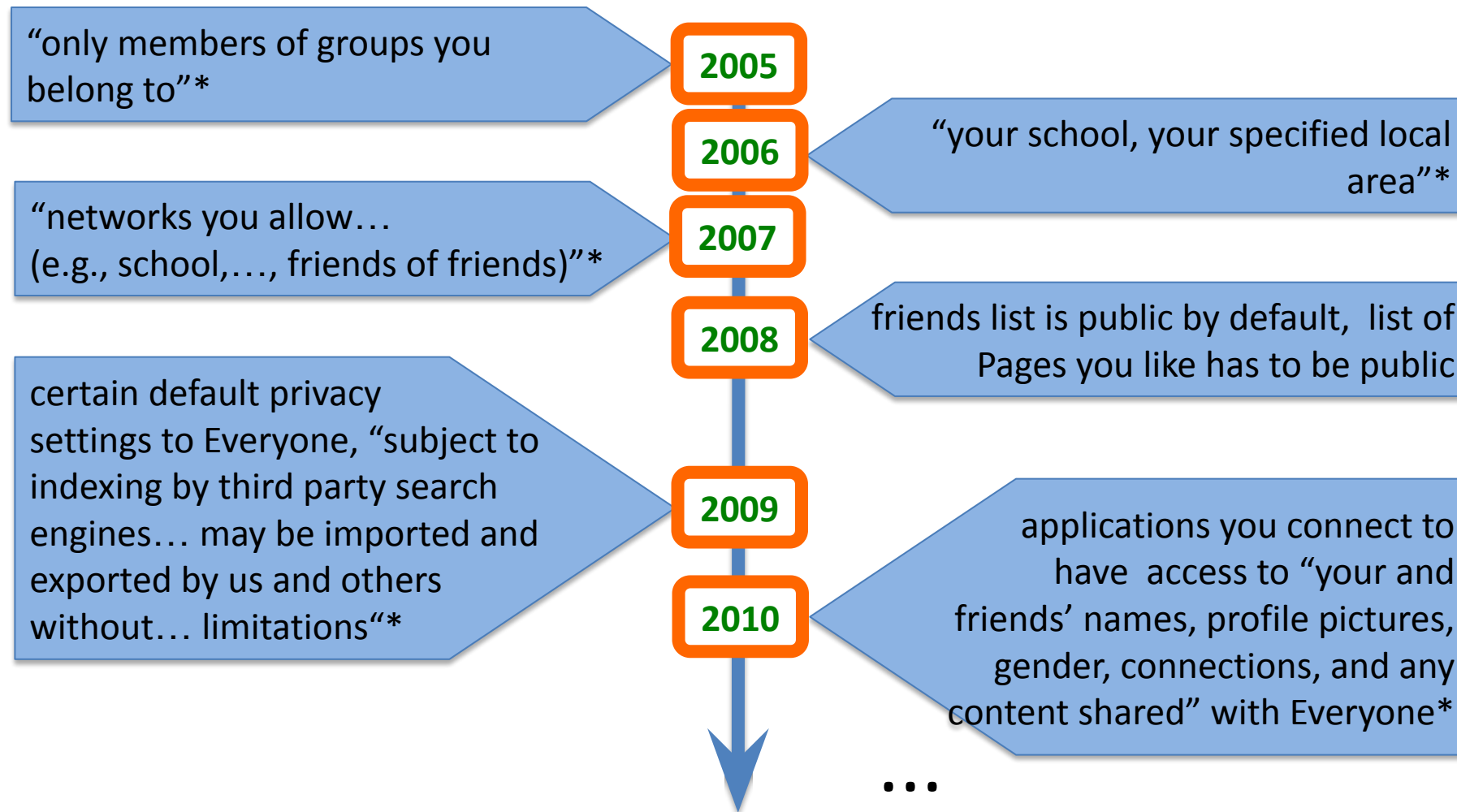
“We’ve lost control of our personal data”

“The current business model for many websites offers **free content in exchange for personal data**. Many of us agree to this – albeit often by accepting **long and confusing terms and conditions** documents – but fundamentally we do not mind some information being collected in exchange for free services. But, we’re missing a trick. As our data is then held in proprietary silos, out of sight to us, we **lose out on the benefits** we could realise if we had direct control over this data, and choose when and with whom to share it. What’s more, **we often do not have any way of feeding back to companies what data we’d rather not share** – especially with third parties – the T&Cs are all or nothing...”

-- Sir Tim Berners-Lee in “Three challenges for the web, according to its inventor,” March 12, 2017



Keeping up with privacy policies



*<https://www.eff.org/deeplinks/2010/04/facebook-timeline>



Personal information leaks in social media

- People reveal private information online
 - Your online profile
 - Birthdate, home town, education, religion, marital status
 - Your Likes and tastes
 - Favorite tv shows, movies, music, food
 - Your status updates
- People reveal private information through their social connections
 - The friends who may reveal private information
 - The groups people join



Privacy in social networks

- **Beyond identity and attribute disclosure**
 - Link re-identification [Zheleva & Getoor '07] (link prediction can reveal sensitive links)
- **Anonymization by removing the profile attributes not enough**
 - Structural re-identification [Backstrom et al. '07]
 - Structural anonymization [Hay et al. '08]
 - Re-identification across networks [Narayanan & Shmatikov '09]
- **Differentially-private analysis of graphs**
 - [Task & Clifton'14, Raskhodnikova & Smith'15]

Privacy in Social Networks. E. Zheleva, E. Terzi, L. Getoor. Morgan & Claypool Publishers 2012.



Privacy in social networks

- Understanding privacy risks
 - Inferring information in social networks [Zheleva '09]
 - Privacy scores [Liu & Terzi '10]
 - Information-sharing models [Kleinberg & Ligett '10]
 - Re-identification through online activity [Kosinski et al, 2014]
- Connecting privacy and fairness
 - FairTest: unwarranted associations framework [Tramer et al. '17]
 - Unintentional vs Intentional decision making

Privacy in Social Networks. E. Zheleva, E. Terzi, L. Getoor. Morgan & Claypool Publishers 2012.



What is private information?

- People may choose **not to reveal** certain pieces of information about their lives
 - Private information includes
 - Sexual orientation, age, income, ...
 - Political affiliation
 - Intelligence
 - **Personality**
 - “the combination of characteristics or qualities that form an individual's distinctive character.”
 - This information might be **predicted**
 - From unrelated publicly disclosed information



Big 5 Personality Traits

- Openness to Experience/Intellect
 - High scorers tend to be original, creative, curious, complex; Low scorers tend to be conventional, down to earth, narrow interests, uncreative.
- Conscientiousness
 - High scorers tend to be reliable, well-organized, self-disciplined, careful; Low scorers tend to be disorganized, undependable, negligent.
- Extraversion
 - High scorers tend to be sociable, friendly, fun loving, talkative; Low scorers tend to be introverted, reserved, inhibited, quiet.



Big 5 Personality Traits

- Agreeableness
 - High scorers tend to be good natured, sympathetic, forgiving, courteous; Low scorers tend to be critical, rude, harsh, callous.
- Neuroticism
 - High scorers tend to be nervous, high-strung, insecure, worrying; Low scorers tend to be calm, relaxed, secure, hardy.



The Big 5 Personality Test

<https://www.truity.com/test/big-five-personality-test>

I see myself as someone who...

1. ...Is talkative

Strongly Disagree 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5 ☐ Strongly Agree

2. ...Tends to find fault with others

Strongly Disagree 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5 ☐ Strongly Agree

3. ...Does a thorough job

Strongly Disagree 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒ Strongly Agree

4. ...Is depressed, blue

Strongly Disagree 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5 ☐ Strongly Agree

5. ...Is original, comes up with new ideas

Strongly Disagree 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒ Strongly Agree

6. ...Is reserved

Strongly Disagree 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5 ☐ Strongly Agree

7. ...Is helpful and unselfish with others



Big 5 Personality Traits

<https://www.truity.com/test/big-five-personality-test>





Predicting personality traits with social media

- Can a user's personality be predicted from his/her publicly available social media profile information?
 - Through self-description, status updates, interests etc. much of the user's personality is reflected
- This paper describes the type of data collected, their methods of analysis and the results of predicting personality traits through machine learning
- Implications to social media design, interface design and other applications such as online marketing and advertising

Golbeck et al (2011) "Predicting Personality Traits with Social Media", in CHI.



Data Collection

Created a Facebook application to administer “Big Five” personality survey to 279 users. Collected their profile statistics:

1. Structural Features

- User’s ego-centric network size, density

2. Personal Information

- user name
- gender
- birthday
- relationship status
- religion
- education
- hometown



Data Collection (cont.)

1. Activities and Preferences

- lists of favorite, such as movies, music etc.

2. Language Features

- This all the text that the users shared through
 - “About Me”, “blurb” text and status updates.
- Of the 279 users, only 167 had enough words to be actually used in this data set.
- Used LIWC to analyze the text producing statistics on 81 different features of text in the 5 categories.



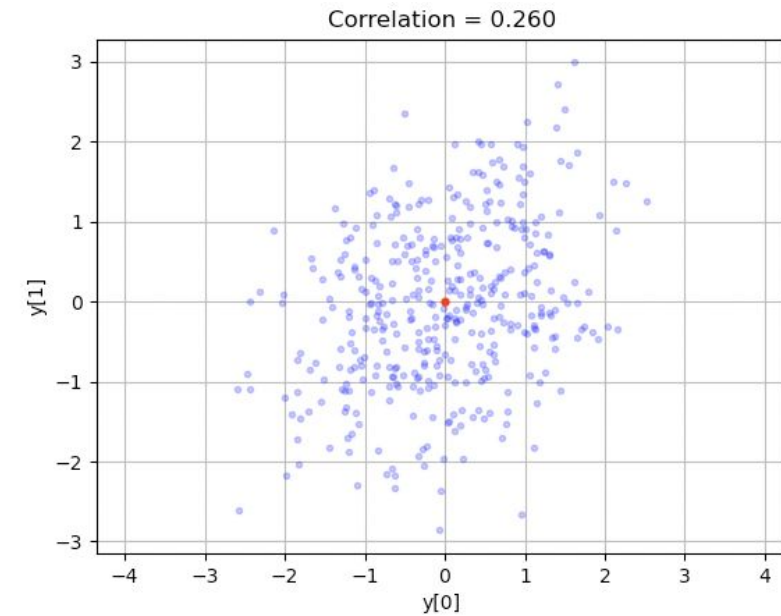
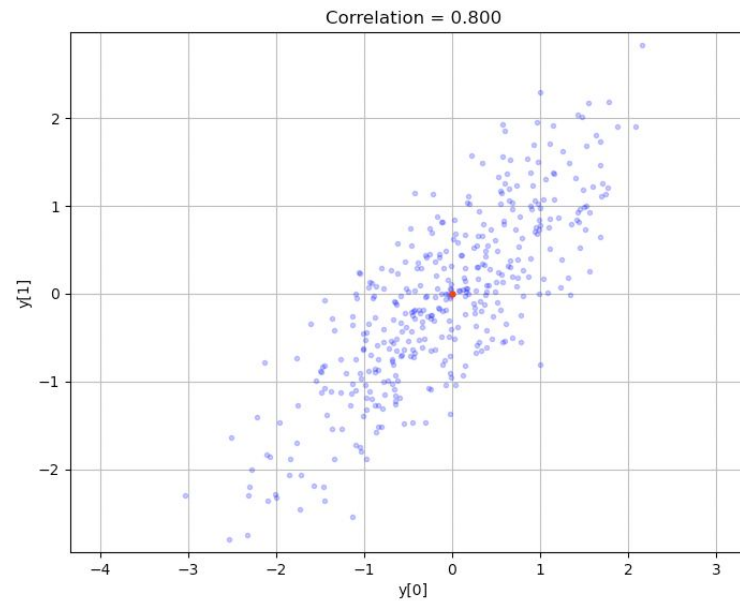
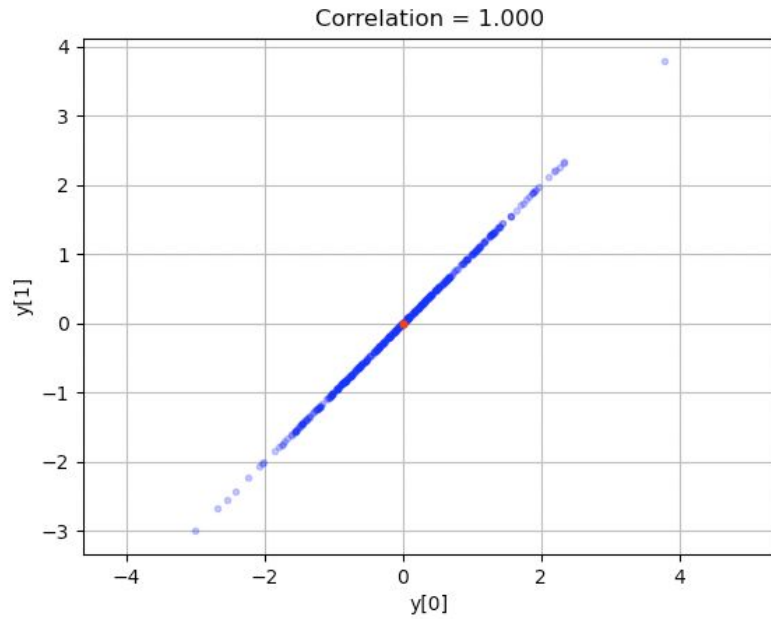
Table 2: Pearson correlation values between feature scores and personality scores. Significant correlations are shown in bold for $p < 0.05$. Only features that correlate significantly with at least one personality trait are shown.

Personality-features correlation

	Open.	Consc.	Extra.	Agree.	Neuro.
Linguistic Features					
Swear Words	0.006	-0.171	0.032	-0.084	-0.120
Social Processes (e.g. Mate, talk, they, child)	0.010	0.264	0.091	-0.022	-0.142
Human Words (e.g. baby, man)	0.078	0.203	0.070	-0.050	-0.062
Affective Processes (e.g. Happy, cried, abandon)	0.105	-0.009	0.136	0.203	0.038
Positive Emotions (e.g. Love, nice, sweet)	0.052	0.045	0.117	0.167	-0.013
Anxiety Words (e.g. Worried, fearful, nervous)	0.044	-0.150	0.008	0.101	0.192
Perceptual Processes (e.g. Observing, heard, feeling)	-0.040	-0.195	-0.163	-0.027	0.096
Seeing Words (e.g. View, saw, seen)	0.060	-0.227	-0.112	0.013	0.067
Biological Processes (e.g. Eat, blood, pain)	-0.014	0.042	0.038	0.154	0.067
Ingestion Words (e.g. Dish, eat, pizza)	-0.098	-0.050	0.029	0.031	0.207
Work Words (e.g. Job, majors, xerox)	0.134	0.096	0.154	0.048	-0.044
Money Words(e.g. Audit, cash, owe)	-0.161	0.024	0.012	-0.006	0.029
Structural Features					
Number of Friends	-0.094	-0.078	0.186	0.013	-0.069
Egocentric Network Density	-0.152	0.050	-0.224	0.059	0.032
Activities and Preferences					
Activities (char length)	0.115	0.095	0.188	0.066	-0.145
Favorite Books (char length)	0.158	-0.093	0.019	0.082	0.028
Personal Information					
Relationship Status (none listed,single, not single)	0.093	0.071	0.194	0.040	-0.036
Last Name length in characters	0.012	-0.111	0.000	-0.044	0.184



Correlation Reminder





Personality and Social Media Profile

Weak correlations between user's profile and personality scores:

- *Conscientiousness*
 - negatively correlated with the frequency of swear words as well as words that described perceptual processes.
 - positively correlated with words concerning social processes.
- *Agreeableness* correlated positively with words describing feelings and
- *Neuroticism* correlated positively with words describing anxiety
- Length of the reported activities and interests correlated with *extraversion* and *openness*

Personality and Social Media Profile



- Extroverts
 - Larger networks (more friends)
 - Less dense network structure.
- Correlation between neuroticism and length of the user's last name.
- Woman were more conscientious, agreeable and neurotic than men.
- Users who chose to share a URL to an external website were more open than those who didn't.



Differences between subpopulations

Value	Open.	Consc.	Extra.	Agree.	Neur.
Male	3.841	3.313	3.145	3.638	2.680
Female	3.671	3.582	3.476	3.806	2.996
<i>p</i>	<i>0.101</i>	<i>0.018</i>	<i>0.018</i>	<i>0.095</i>	<i>0.018</i>
No Website	3.710	3.495	3.264	3.697	2.900
Website	4.010	3.498	3.508	3.773	2.770
<i>p</i>	<i>0.003</i>	<i>0.978</i>	<i>0.071</i>	<i>0.382</i>	<i>0.275</i>



Predicting Personality

- Predicted personality trait score based on 74 user features
 - using regression with 10-fold cross-validation
 - The Mean Absolute Prediction Error around 11%

Factor	M5'Rules	Gaussian	M5'Rules	Gaussian
	MAE		Correlation Coefficient	
Open.	0.099	0.117	0.653	0.179
Consc.	0.104	0.117	0.595	0.094
Extra.	0.138	0.124	0.553	0.050
Agree.	0.109	0.117	0.482	0.150
Neuro.	0.127	0.117	0.531	0.106



Predicting private traits from digital records of human behavior

[Kosinski, M., Stillwell, D., & Graepel, T. (2013). Private traits and attributes are predictable from digital records of human behavior. *PNAS*]

- Facebook users employ 'likes' to express their positive attitudes towards online content and entities
 - Publicly available
 - Similar to Web search queries, Web browsing history, and credit card purchases.
- Statistical method to automatically and accurately estimate personal attributes from Facebook 'likes'



Simon's Likes on Sports



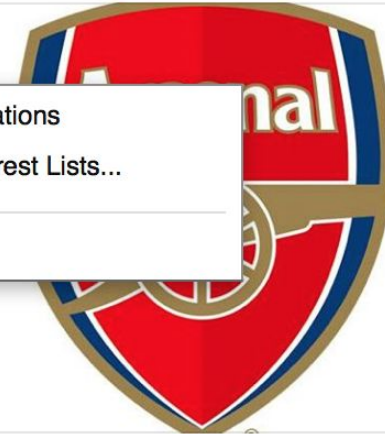
FC Barcelona ✓



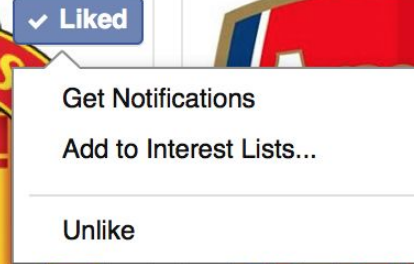
Chelsea Football Club ✓



Manchester United ✓



Arsenal ✓



Los Angeles Dodgers ✓

Add Teams

Choose from your suggestions or search for teams you like.

What kinds of sports does Simon like?



Simon's Likes on Music

Can you guess Simon's music tastes ?



Michael Jackson ✓
Musician/Band



Adele ✓
Musician/Band



David Guetta ✓
Musician/Band



Eminem ✓
Bands & Musicians



제이레빗 (J Rabbit)
Musician/Band



타루 - Taru
Musician/Band



Moniker
Musician/Band

Add Music

Choose from your suggestions or
search by artist.



Get the Citi Mobile® App
banking.citibank.com

With Citi Mobile, you can pay
people, deposit checks and
more. With just your thumbs.



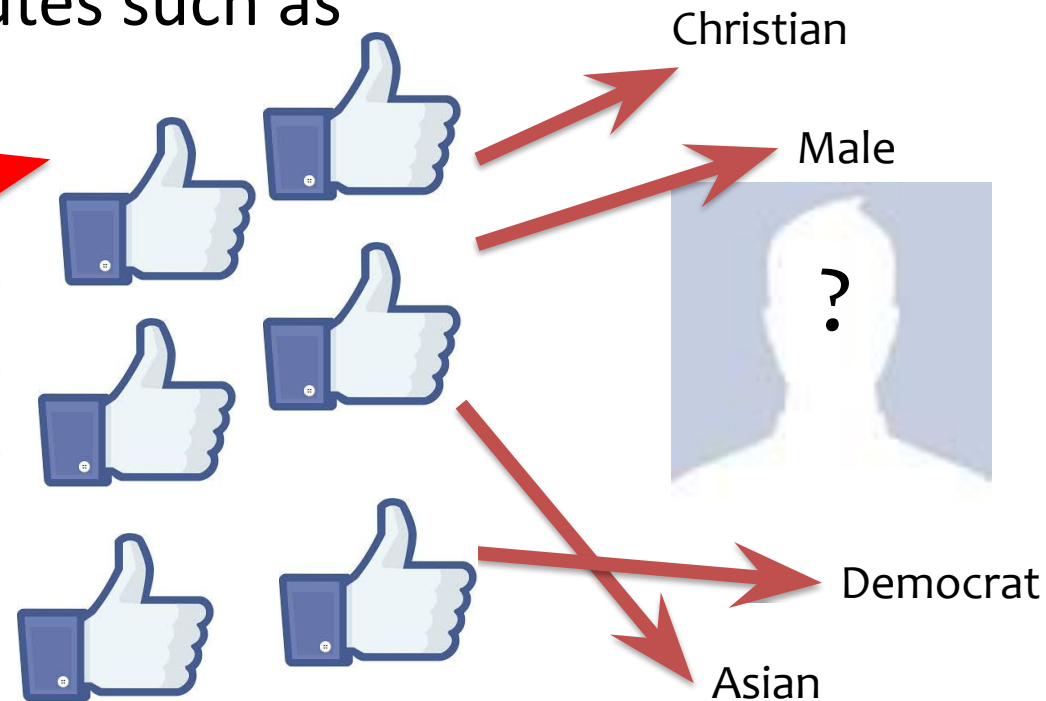
Marriott Rewards faster
creditcards.chase.com
Plus earn 5 points per \$1
spent at Marriott(R) locations
worldwide. Learn more.

Facebook Likes -> Personal Traits (Digital Fingerprinting)



Likes in Facebook can be used to automatically and accurately predict with high probability for the highly sensitive personal attributes such as

- gender
- ethnicity
- religious
- political views
- personal traits
- intelligence
- happiness





Main Questions

- Can we ***automatically*** estimate a wide range of personal attributes that people would typically consider to be private ?
- Can we ***accurately*** estimate a wide range of personal attributes that people would typically assume to be private ?
- Can we use only **Facebook Likes** ?



Design of User Study

- **Step1.** Selected the private and intrusive attributes
 - sexual orientation, ethnicity, political views, religion
 - personality, intelligence, satisfaction with life (SWL), substance use (alcohol, drugs, cigarettes), whether an individual's parents stayed together until the individual was 21 years old
 - basic demographic attributes such as age, gender, relationship status, and size and density of the friendship network



Design of User Study

- **Step2.** Data Collection

They obtained data set from 58,000 volunteers who share their 10M Facebook Likes, profiles, and took psychometric surveys.



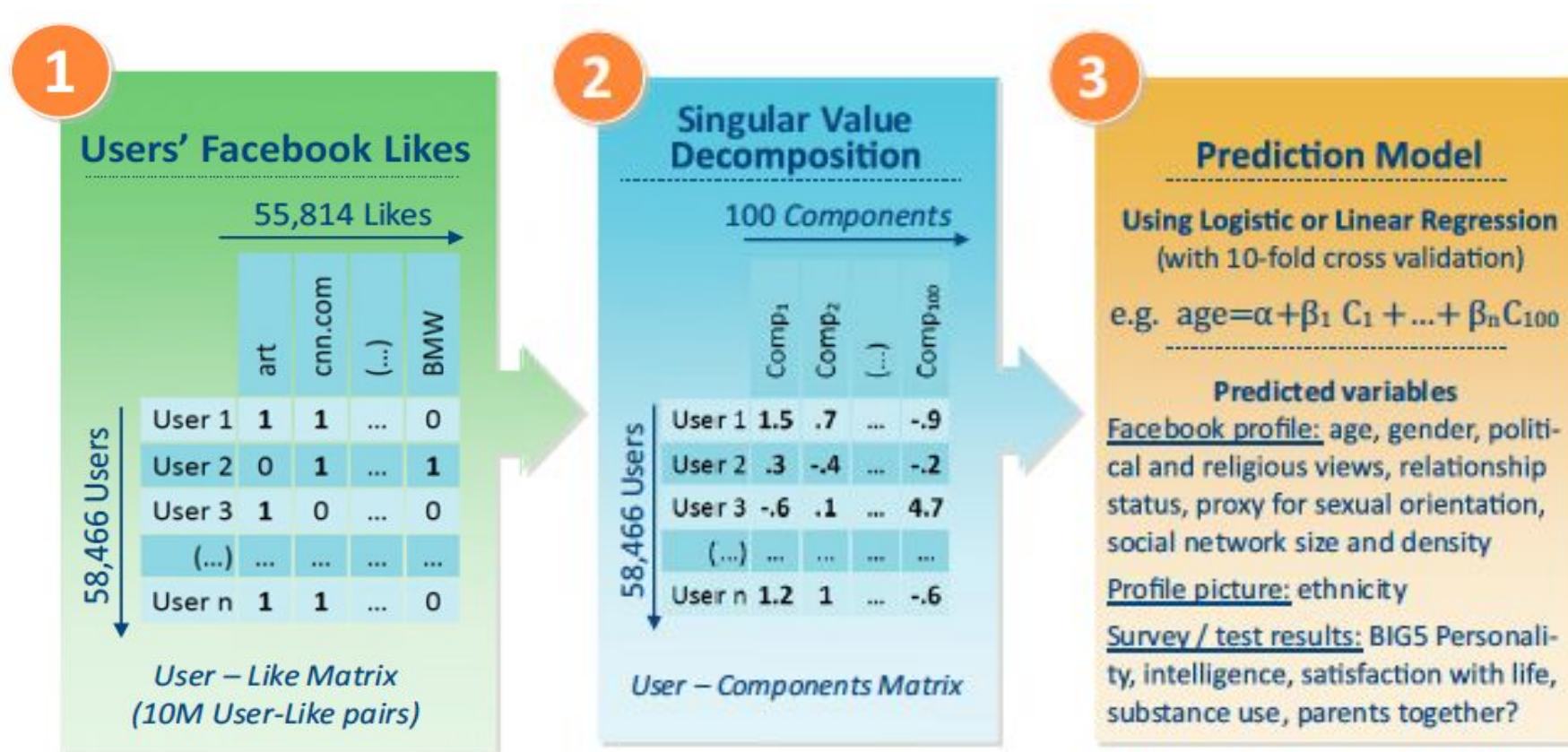


Methodology/Models

- **Truncated SVD for computation and data analysis**
 - Used top 30~100 SVD components, depending on the attributes.
- **Logistic/linear regression model for Prediction**
 - $\text{Gender} = \alpha_{\text{Gender}} + \beta_1 C_1 + \beta_2 C_2 + \dots + \beta_i C_i + \dots + \beta_{30} C_{30}$
 - $\text{Age} = \alpha_{\text{Age}} + \beta_1 C_1 + \beta_2 C_2 + \dots + \beta_i C_i + \dots + \beta_{100} C_{100}$
- **Evaluations**
 - Pearson product–moment correlation coefficient between the actual and predicted values to measure the accuracy of prediction.

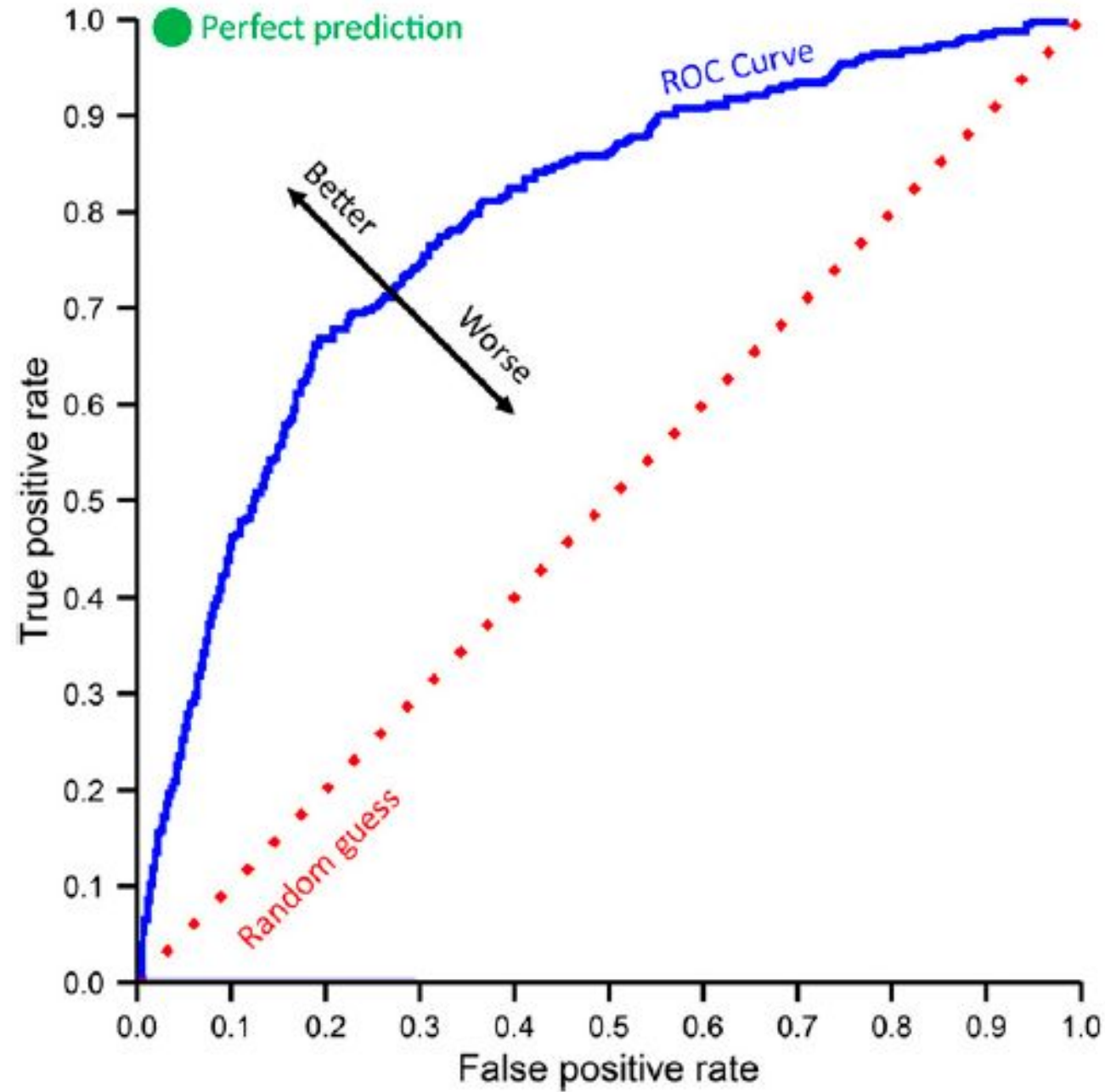


Prediction with Facebook likes



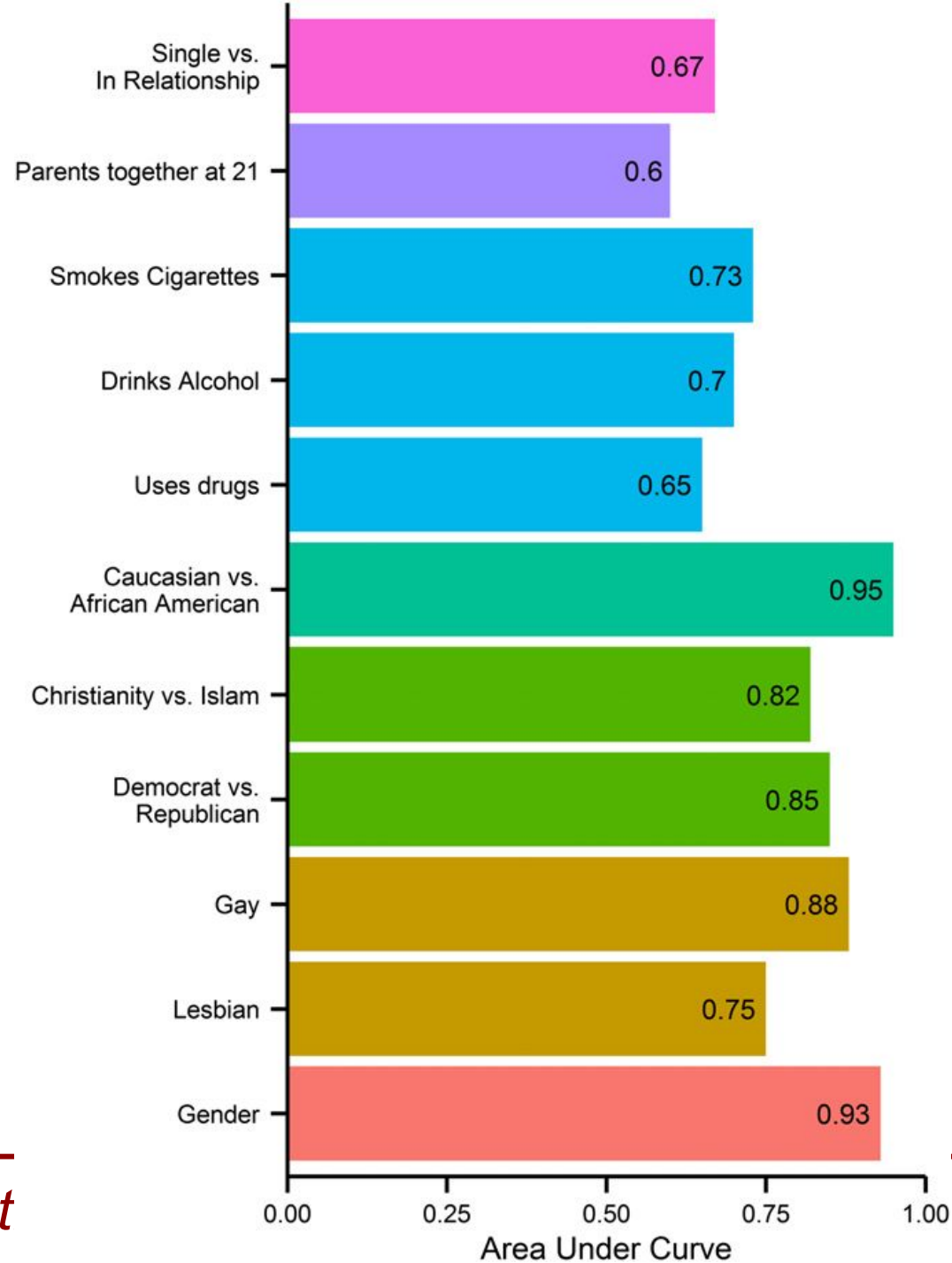


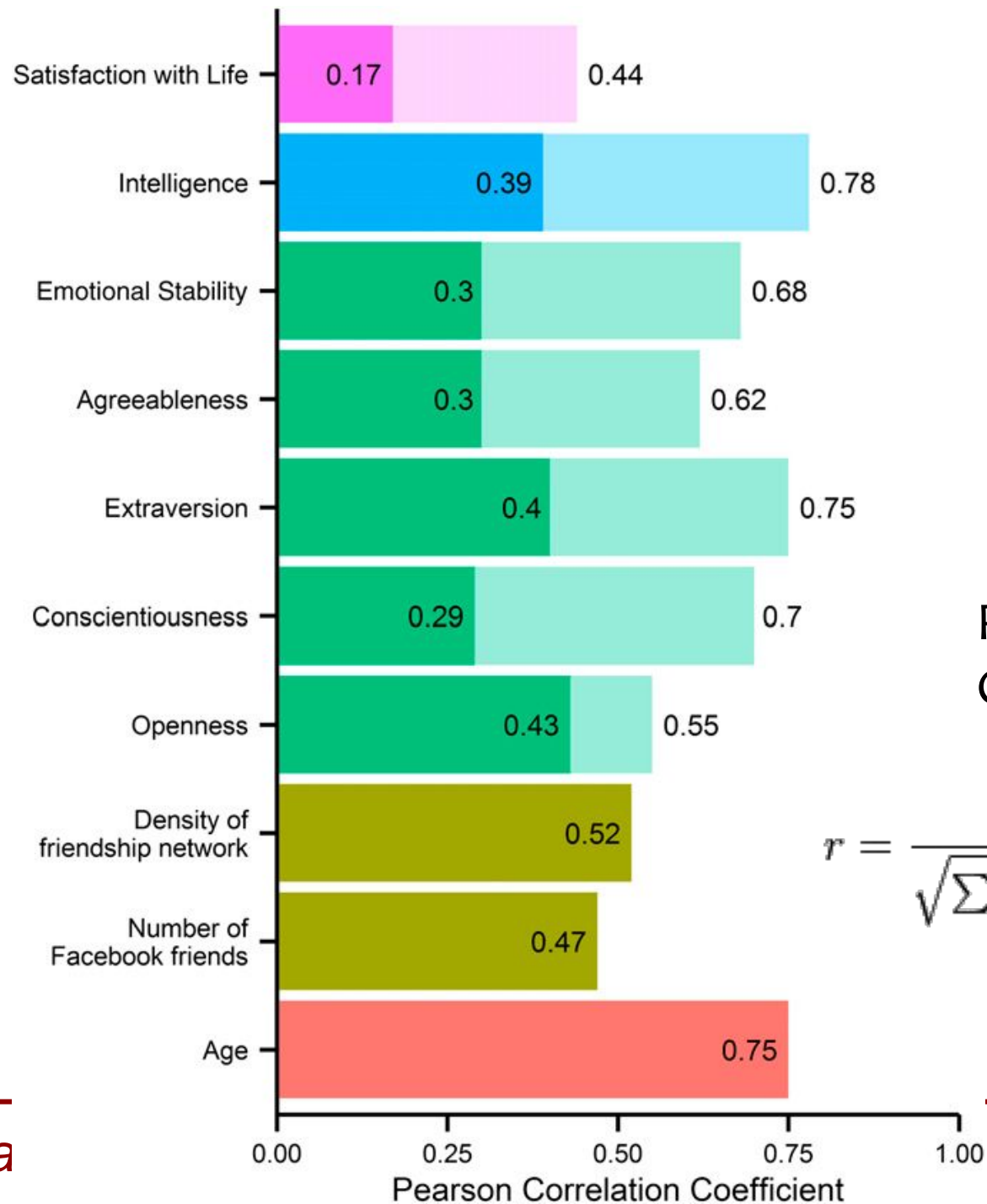
ROC Curve





Classification accuracy for binary personal attributes



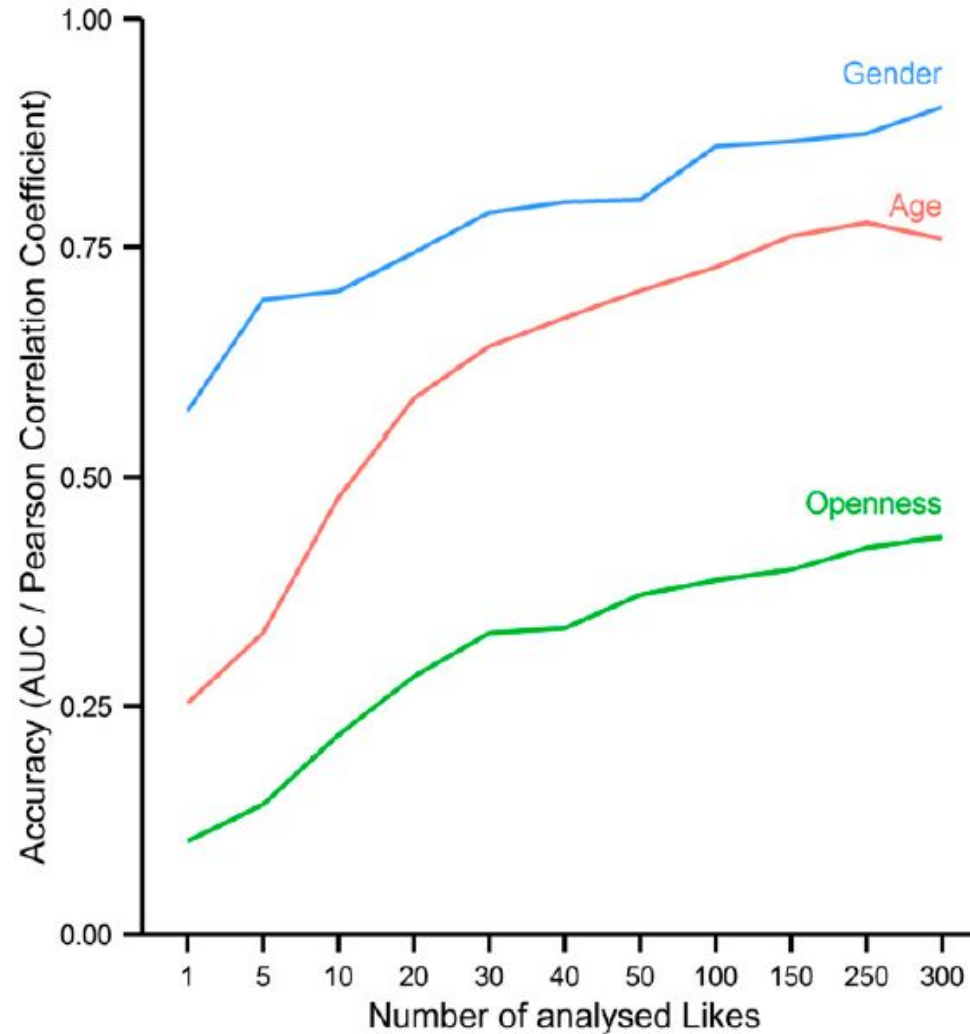


Accuracy of regression
for numeric attributes

Pearson Correlation
Coefficient r

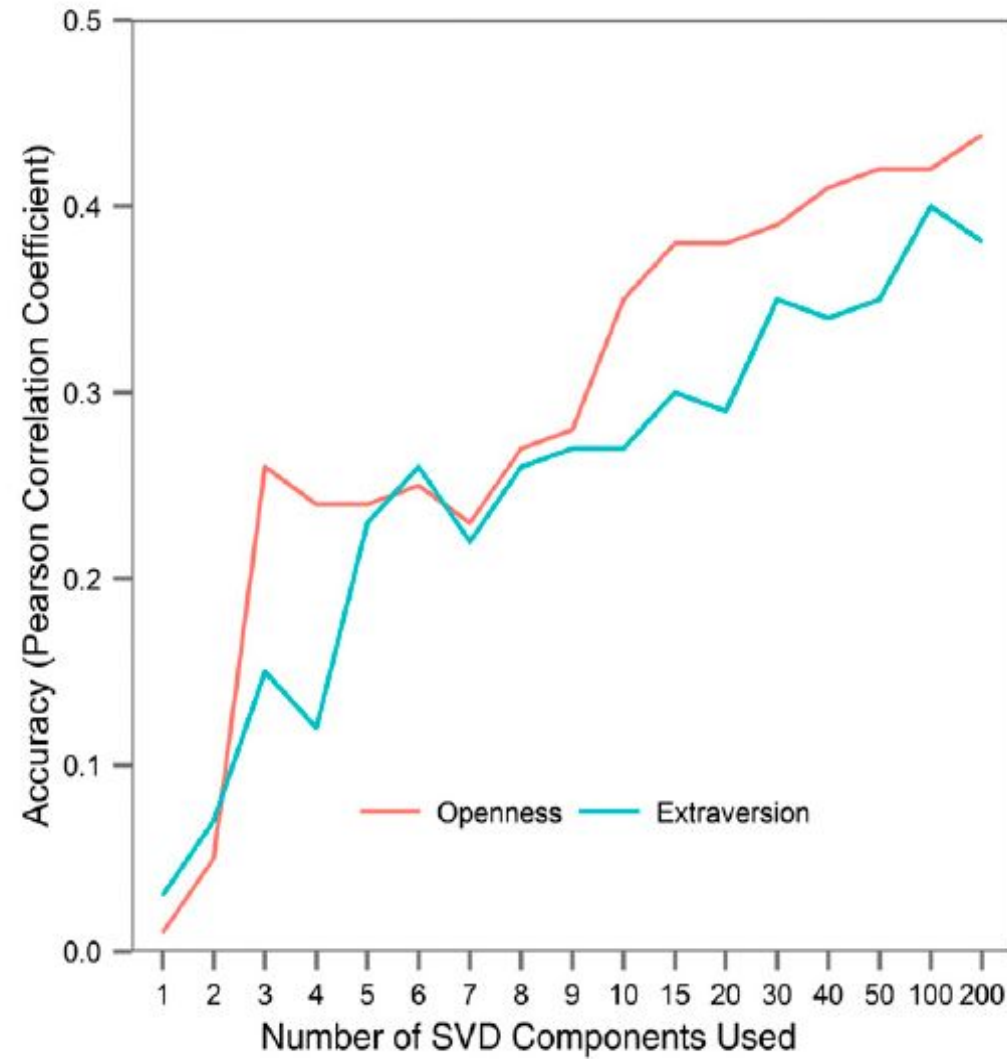
$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

Prediction accuracy vs volume of data (#likes)





Prediction accuracy vs #SVD components





Most predictive likes

IQ	High	Low
	The Godfather Mozart Lord Of The Rings	Sephora I Love Being A Mom Harley Davidson
Extraversion	Outgoing & Active	Shy & Reserved
	Beerpong Dancing Cheerleading Chris Tuger Theatre	Video Games Programming Role Playing Games Manga Minecraft
Openness	Liberal & Artistic	Conservative
	Oscar Wilde Leonardo Da Vinci Leonard Cohen Bauhaus	NASCAR ESPN2 The Bachelor Justin Moore



Smoking	Yes	Cradle Of Filth Under Armour Slayer Band Inbox 1 Makes Me Nervous Dimebag Darrell Rob Zombie I Always Accept The Terms And Conditions Without Reading Them I Bottle Everything Up Until I Finally Snap Life Is Better In Summer Screwing Around In Walmart	That Spider Is More Scared Than U Are Oh Really Did It Tell U That Honda Move Out Of The Way Children I've Been Waiting 11 Years To See Toy Story 3 FBI Open The Door No Its Cool When You Break In How To Make A Girl Smile <3<3 The Desk Able To Protect You From Fire Earthquakes And Nuclear War When Your Fortune Cookie Knows What's Up Rocky When Little Kids Are Chasing Me I Run Slow So They Think They're Fast I Drop My I-pod Then My Headphones Save Its Life	No



Age	<i>Old</i>	Cup Of Joe For A Joe Coffee Party Movement Dr Mehmet Oz Fixit And Forgetit The Closer Joyce Meyer Ministries Proud To Be A Mom Freedomworks Small Business Saturday Fly The American Flag	Walt Disney Records Body By Milk Harperteen J Bigga Because I Am A Girl I Hate My Id Photo 293 Things To Do In Class When You Are Bored Dude Wait What JCP Teen	<i>Young</i>
	<i>Many</i>	Mojo-Jojo Biology Dollar General Hillary 106 & Park Jennifer Lopez Paid In Full Yo Gotti The Dollar You Are Holding Could've Been In A Stripper's Butt Crack	The Dark Knight In'n'out Burger Hard Rock Honey, Where Is My Supersuit Hating ICP Minecraft Iron Maiden Walking With Your Friend & Randomly Pushing Them Into Someone/Something	<i>Few</i>

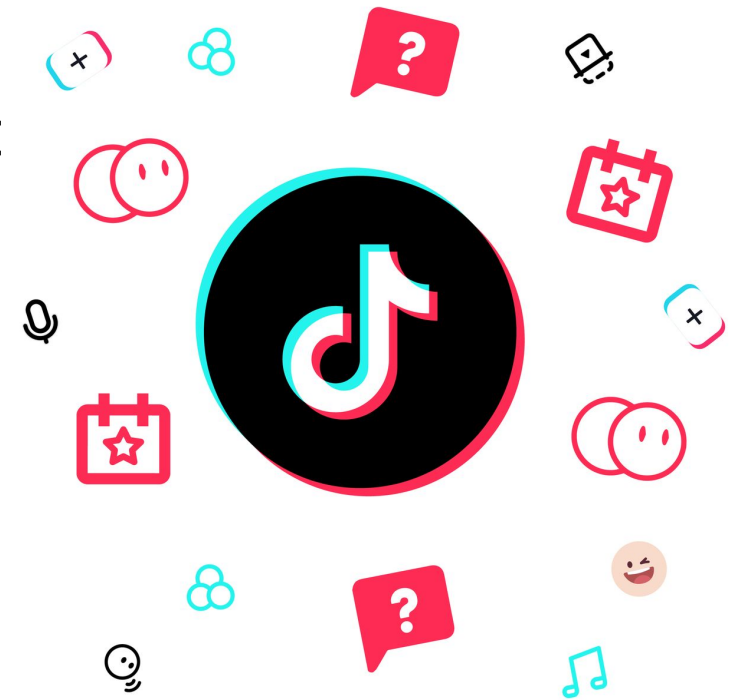


Agreeableness	<i>Cooperative</i>	Compassion International	I Hate Everyone	<i>Competitive</i>
		Logan Utah	I Hate You	
		Jon Foreman	I Hate Police	
		Redeeming Love	Friedrich Nietzsche	
		Pornography Harms	Timmy South Park	
		The Book Of Mormon	Atheism / Satanism	
		Circles Of Prayer	Prada	
		Go To Church	Sun Tzu	
		Christianity	Julius Caesar	
		Marianne Williamson	Knives	
Emotional Stability	<i>Neurotic</i>	Sometimes I Hate Myself	Business Administration	<i>Calm & Relaxed</i>
		Emo	Getting Money	
		Girl Interrupted	Parkour	
		So So Happy	Track & Field	
		The Addams Family	Skydiving	
		Vocaloid	Mountain Biking	
		Sixbillionsecrets.com	Soccer	
		Vampires Everywhere	Climbing	
		Kurt Donald Cobain	Physics / Engineering	
		Dot Dot Curve	48 Laws Of Power	
		To Enactio	Modern Warfare 2	



TikTok

- TikTok “reads your mind”: algorithm takes your data and learns about you, including your psychology.
- TikTok analyzes how long users watch different videos to learn personal traits like musical tastes, political views, sexual orientation, or signs of mental health issues, often without users realizing it.
- It tailors your page to content it thinks you’ll enjoy.
- Its *ultimate goal* is to maximize **daily active users** by optimizing two key metrics: **retention** (whether users return) and **time spent** on the app.



<https://www.nytimes.com/2021/12/05/business/media/tiktok-algorithm.html>



Conclusions

- Many private attributes can be inferred from people's publicly accessible activity, such as Facebook likes
 - Some attributes are remarkably easy to predict... even a single like can lead to significant improvement over random prediction
- This can have benefits, but also risks with respect to privacy

How Trump Consultants Exploited the Facebook Data of Millions

By Matthew Rosenberg, Nicholas Confessore and Carole Cadwalladr

March 17, 2018



[Leer en español](#)

(After this story was published, Facebook came under harsh criticism from lawmakers in the United States and Britain. [Read the latest.](#))

LONDON — As the upstart voter-profiling company [Cambridge Analytica](#) prepared to wade into the 2014 American midterm elections, it had a problem.

Facebook Halts Ad Targeting Cited in Bias Complaints

By Noam Scheiber and Mike Isaac March 19, 2019

After years of criticism, Facebook announced on Tuesday that it would stop allowing advertisers in key categories to show their messages only to people of a certain race, gender or age group.

The company said that anyone advertising housing, jobs or credit — three areas where federal law prohibits discrimination in ads — would no longer have the option of explicitly aiming ads at people on the basis of those characteristics.

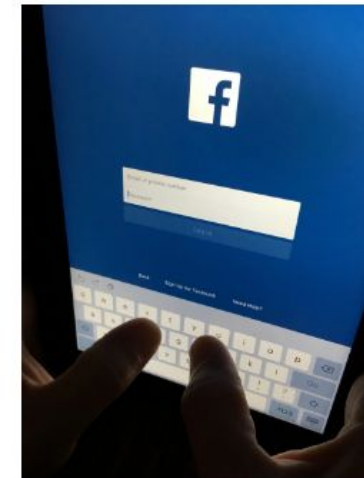
Facebook and Cambridge Analytica: What You Need to Know as Fallout Widens

By Kevin Granville

March 19, 2018



[Leer en español](#)



Cambridge Analytica, a political data firm hired by President Trump's 2016 election campaign, gained access to information on 50 million Facebook users as a way to identify the personalities of American voters and influence their behavior.

Elise Amendola/Associated Press



How Facebook Enabled Psychological Targeting

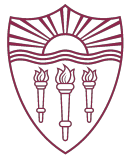
- The **Cambridge Analytica scandal** revealed how personal data harvested from Facebook was used to create psychological profiles of millions of users without their consent — and how these profiles were exploited to target them with highly personalized political ads.
 - In 2014, an academic researcher created a Facebook quiz ("thisisyourdigitallife") that collected data from people and their **Facebook friends** — reaching around **87 million users**.
 - Users believed the quiz was for academic research, but data was sold to **Cambridge Analytica**, a political consulting firm
 - Cambridge Analytica used the data to build detailed **psychological profiles** of users — estimating personality traits
 - Used the profiles to **microtarget** users with highly personalized political ads designed to exploit their specific fears, emotions, and motivations.
 - For example, users identified as anxious and conservative were shown ads emphasizing immigration dangers or law-and-order messages
 - used in major political campaigns, including **Brexit** and the **2016 U.S. presidential election**.

Cambridge Analytica scandal

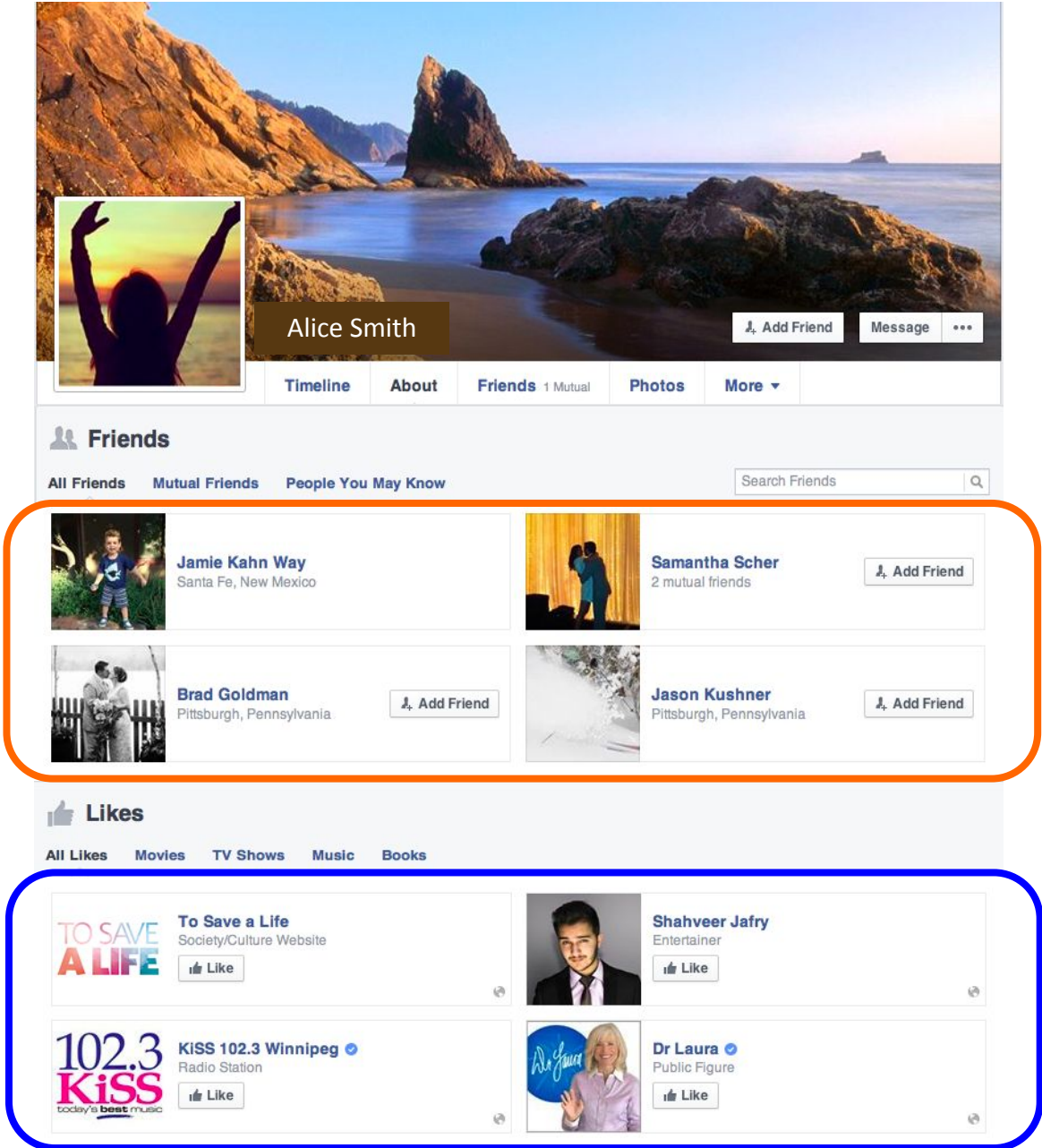


- Why was this a scandal
 - **Consent violation:** Users never gave informed consent for their data — or their friends' data — to be used for political profiling.
 - **Manipulation:** Psychological targeting enabled **emotional manipulation** on a mass scale, steering people's decisions without their full awareness.
 - The scandal severely damaged public trust in Facebook, leading to hearings in Congress, major fines (including a **\$5 billion** FTC fine), and global conversations about **data privacy and algorithmic influence**.
 - powerful data-driven targeting can be used not just to persuade, but to **exploit psychological vulnerabilities**.

Privacy in social networks



Social
links



Group
affiliation
s

[slides courtesy of Elena Zheleva]



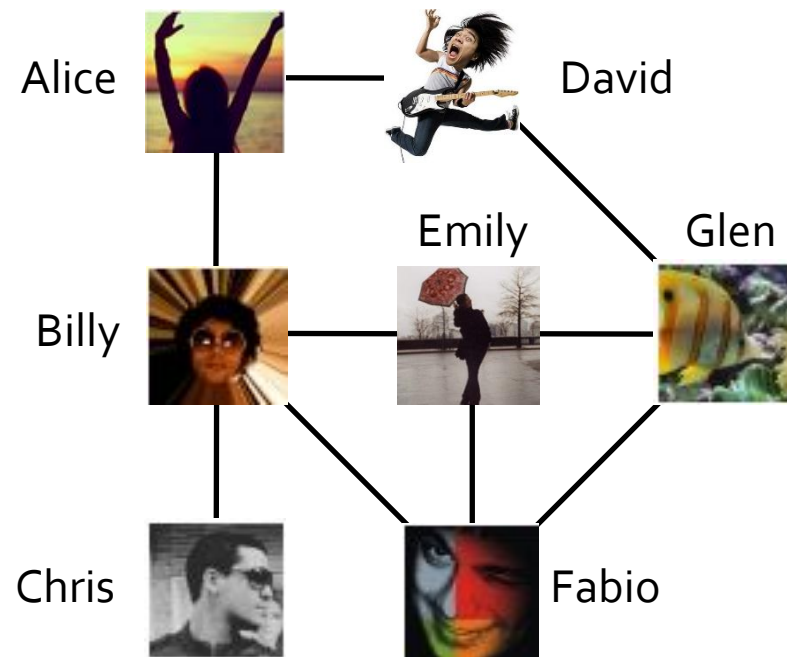
- Group affiliations often cannot be hidden!
- Hypothesis: you can predict private attributes based on public information
- Assumptions:
 - An online social network with public AND private profiles
 - Social links and group affiliations are always public
 - Adversary's goal: to predict attributes in private profiles
 - Adversary can build probabilistic models using public data



From networks to graphs

Social network

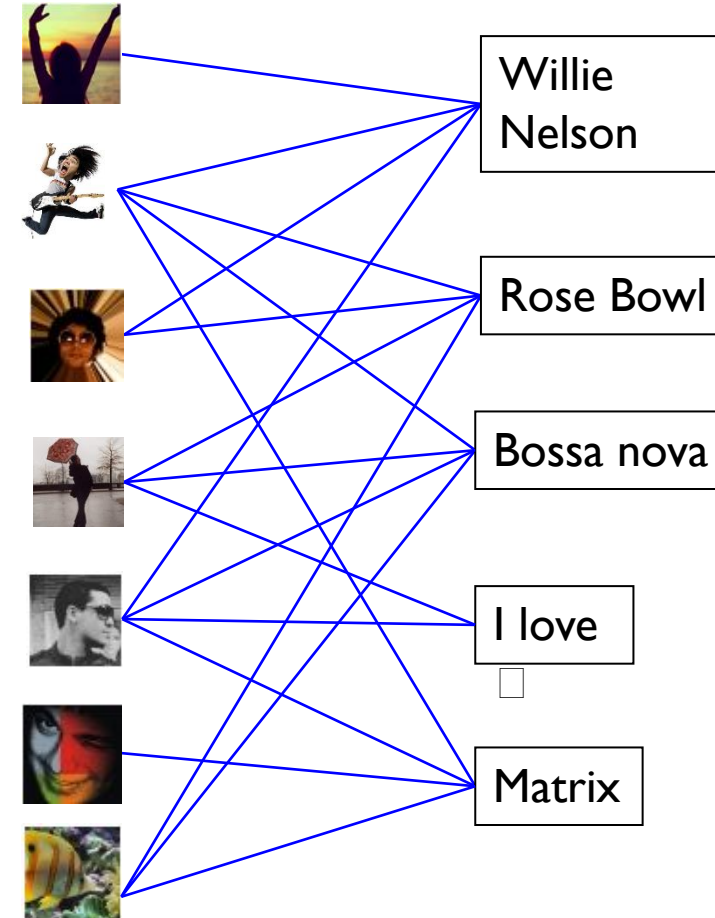
w/ friendship links



- Large, high-dimensional
- Multi-modal
- Multi-relational

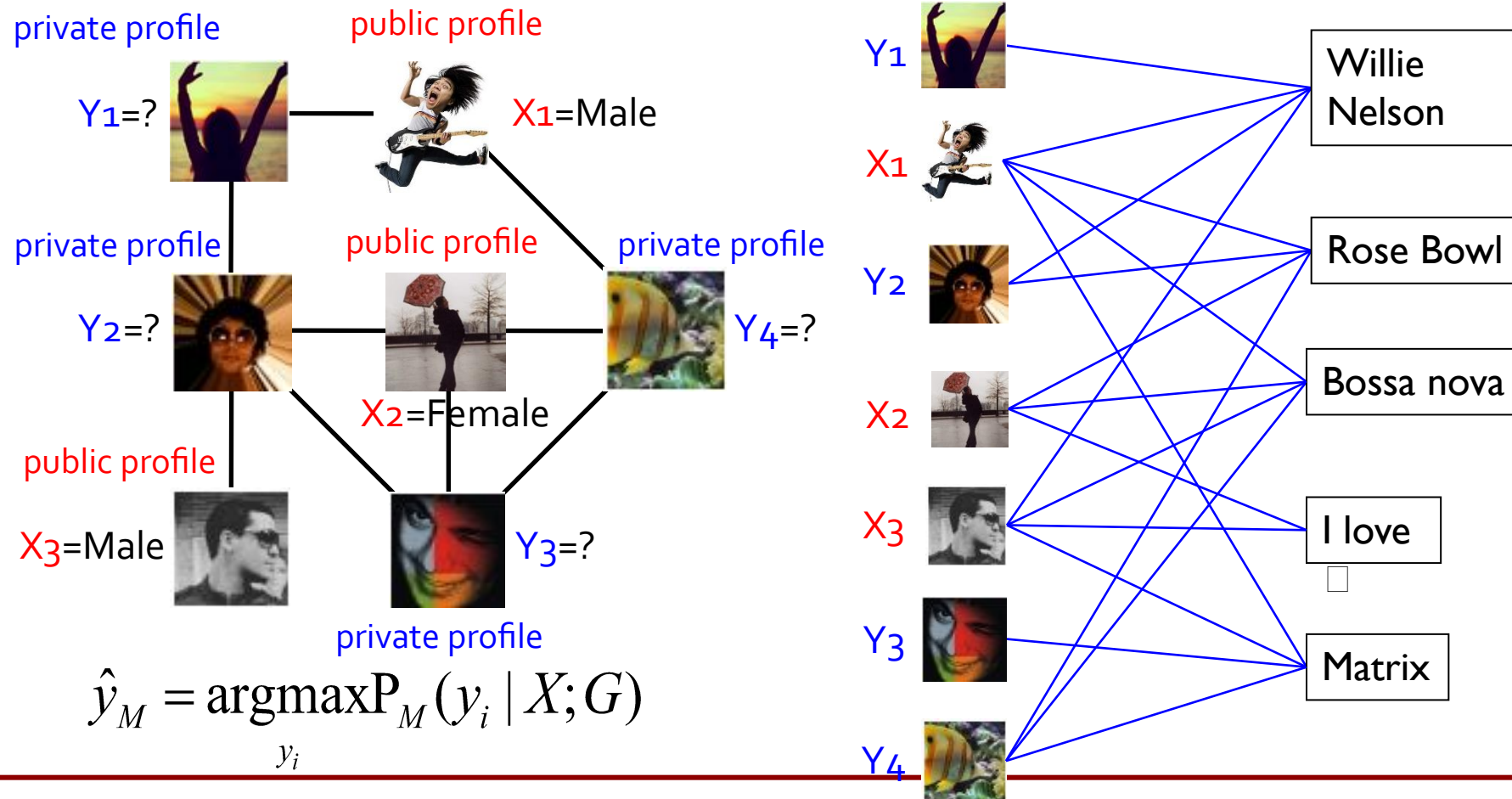
Affiliation network

w/ group links



Sensitive attribute disclosure

- If an adversary is able to determine the value of a user attribute that the user intended to stay private



Collective classification*

Given a set of trait labels (e.g., liberal or not), label the other objects in the network accurately

Is



liberal?

No links

Traditional ML: independent traits

Gender	Location	Liberal
Male	DC	Yes
Female	DC	?



Social links

Friends $\leftrightarrow$ common trait

Alice



David

Interest group links

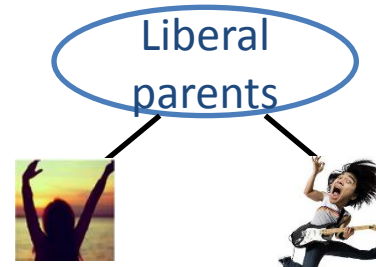
Affiliation $\leftrightarrow$ common trait



Willie
Nelson

Latent links

Trait due to common unobserved factor

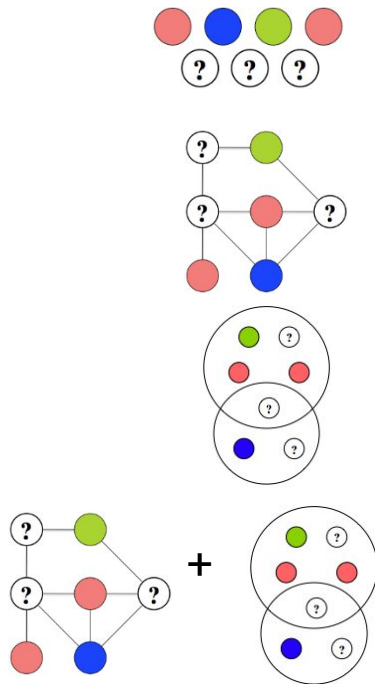


**Collective Classification in Network Data*. Sen et al. AI Magazine 2008.



Collective classification

Assume adversary can apply a probabilistic model M to predict the sensitive attribute



– **No links:** BASIC

- Majority label

– **Social links:** AGG, CC, BLOCK, LINK

- Feature space size: number of users

– **Group links:** CLIQUE, GROUP, GROUP*

- Feature space size: number of groups

– **Both link types:** LINK-GROUP

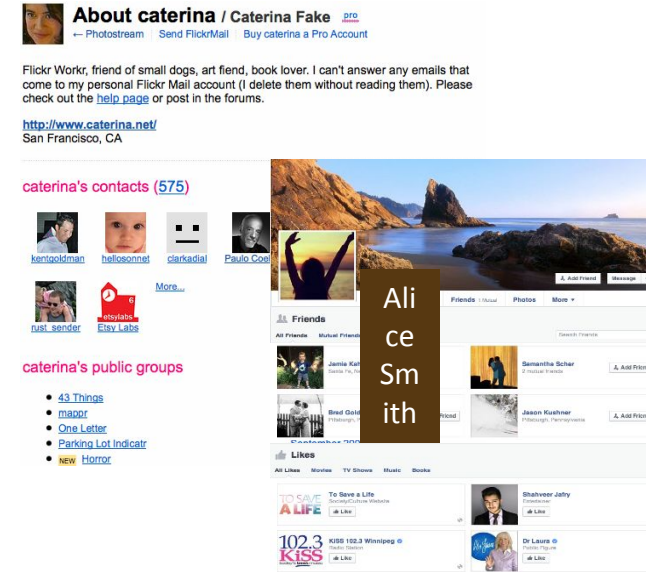
- Feature space size: number of users and groups

[Zheleva, Getoor. (2009) To Join or not to Join: the Illusion of Privacy in Social Networks with Mixed Public and Private User Profiles. In WWW.]



Data and experimental setup

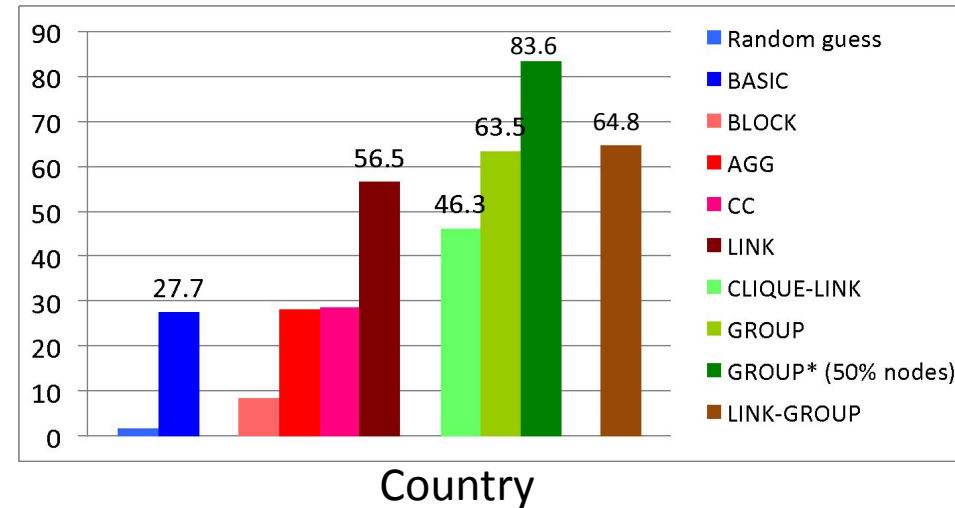
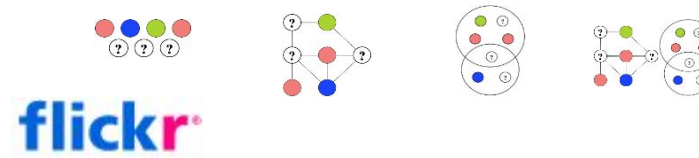
- Flickr: snowball sample
 - ~9,000 profiles, 1 million links, 50,000 groups
 - sensitive: location (55 values)
- Facebook: all freshmen (Harvard)
 - ~1,600 profiles, 86,000 links, 3,000 groups
 - sensitive: gender (2) and political views (6)
- Assign each profile to be public with probability=50%
 - Train model on public profile data (10 different models)
 - Predict on private profiles



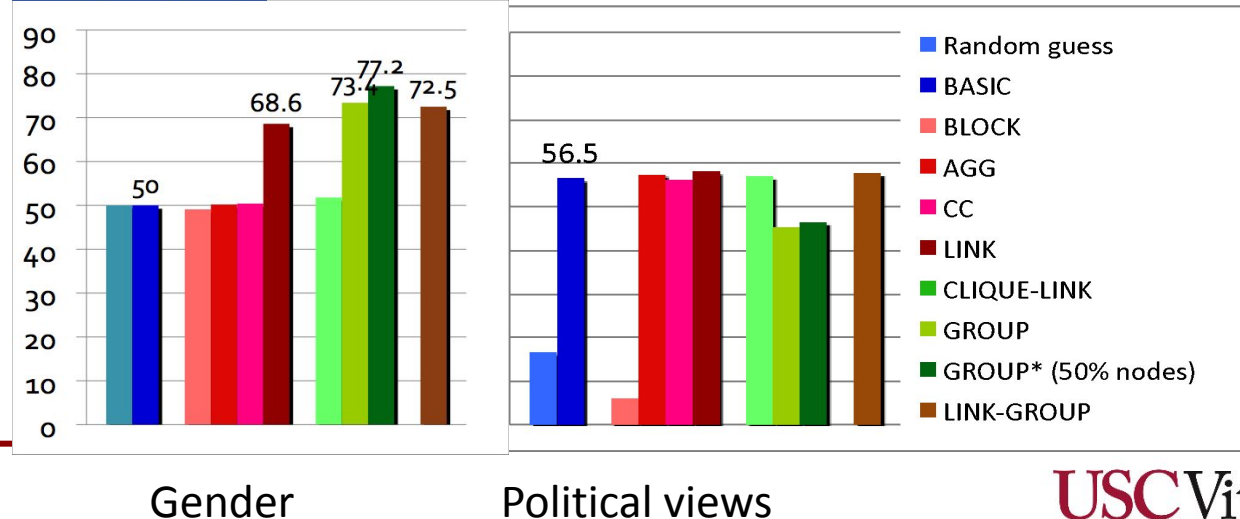


Findings

- Affiliation networks have a high potential for leaking private information
 - More informative than friendship network
 - Homogeneous groups most informative



facebook

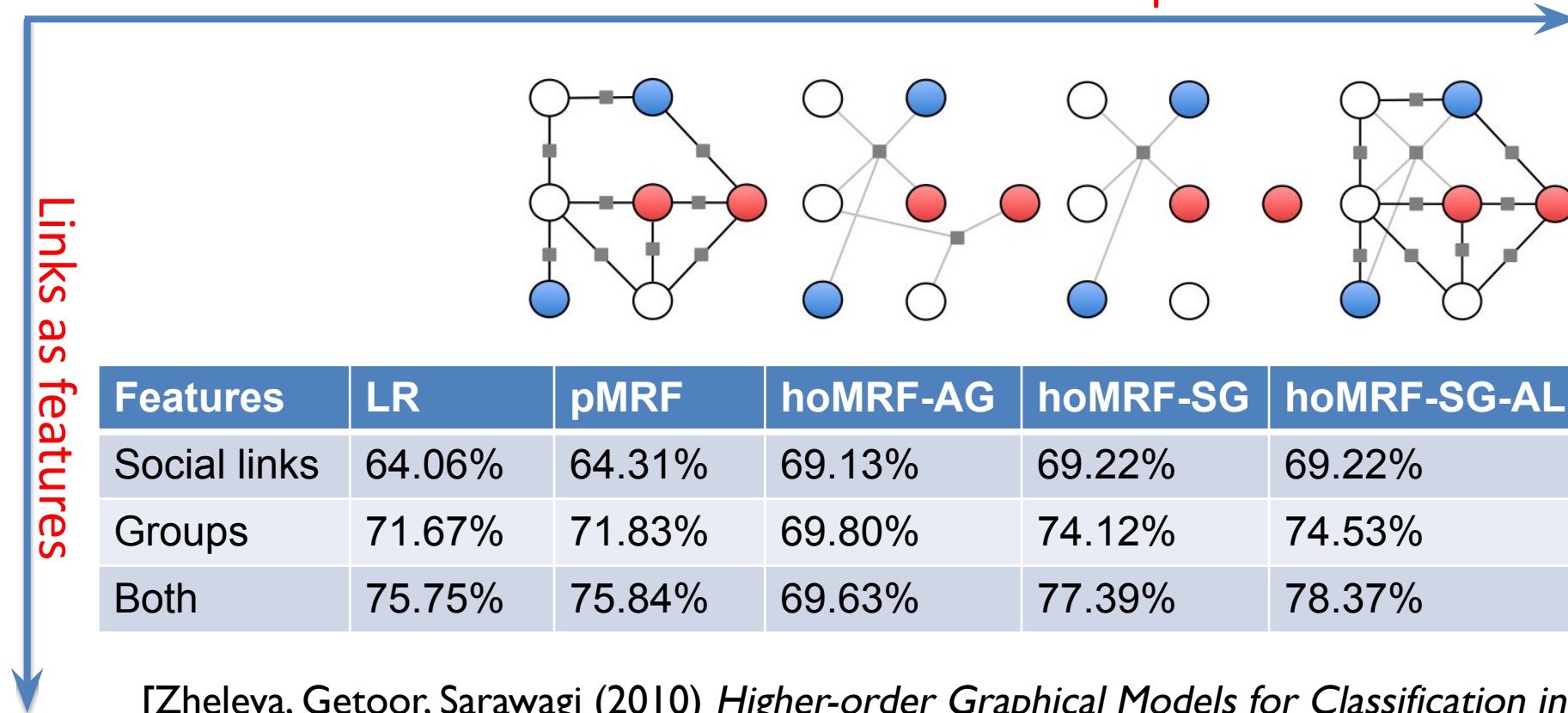




Increasing model complexity

- Predict “gender” in the Facebook dataset

Links as structural statistical dependencies



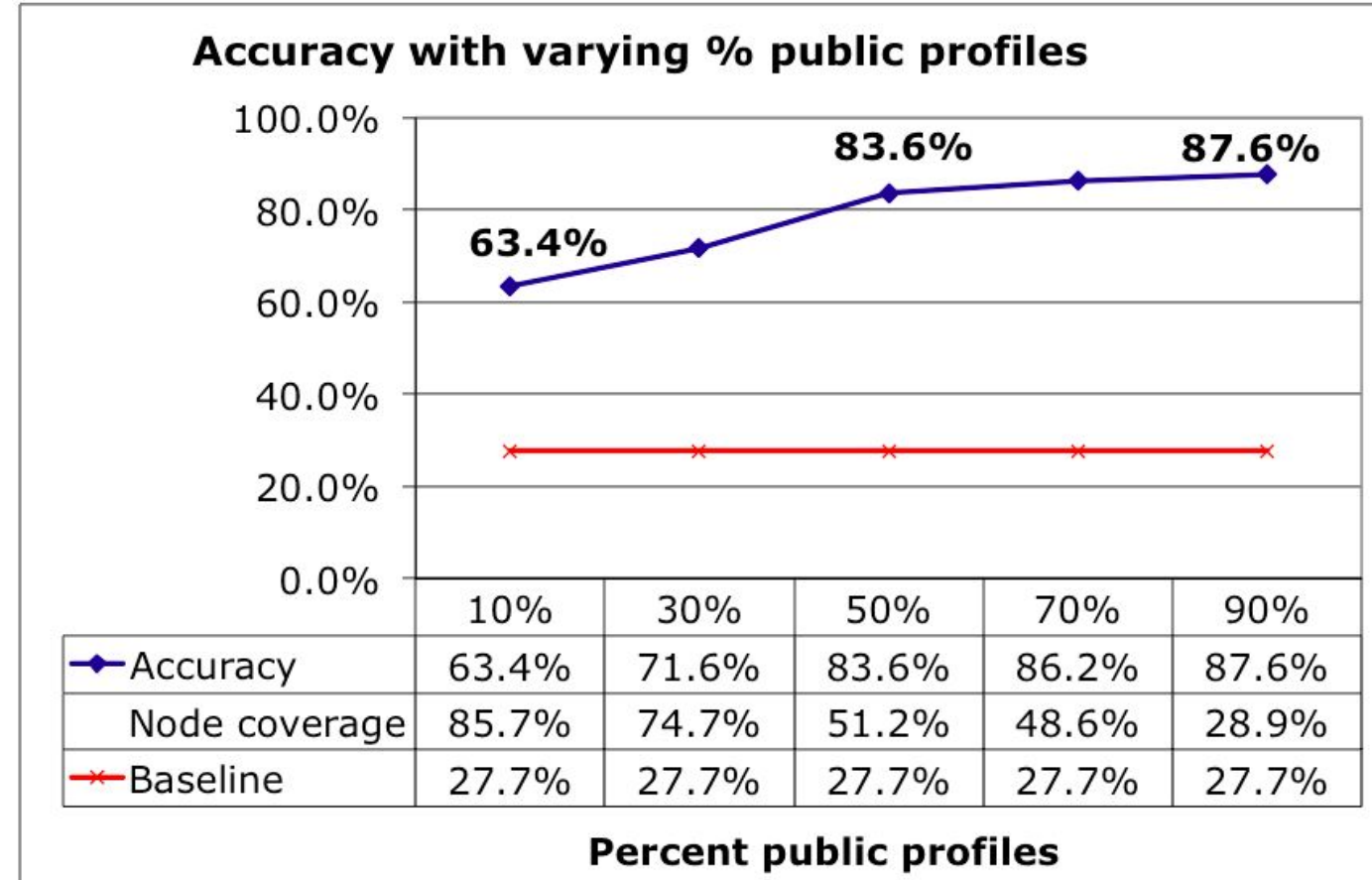
Features	LR	pMRF	hoMRF-AG	hoMRF-SG	hoMRF-SG-AL
Social links	64.06%	64.31%	69.13%	69.22%	69.22%
Groups	71.67%	71.83%	69.80%	74.12%	74.53%
Both	75.75%	75.84%	69.63%	77.39%	78.37%

[Zheleva, Getoor, Sarawagi (2010) *Higher-order Graphical Models for Classification in Social and Affiliation Networks*. NIPS Workshop on Networks Across Disciplines.]



Your privacy depends on others

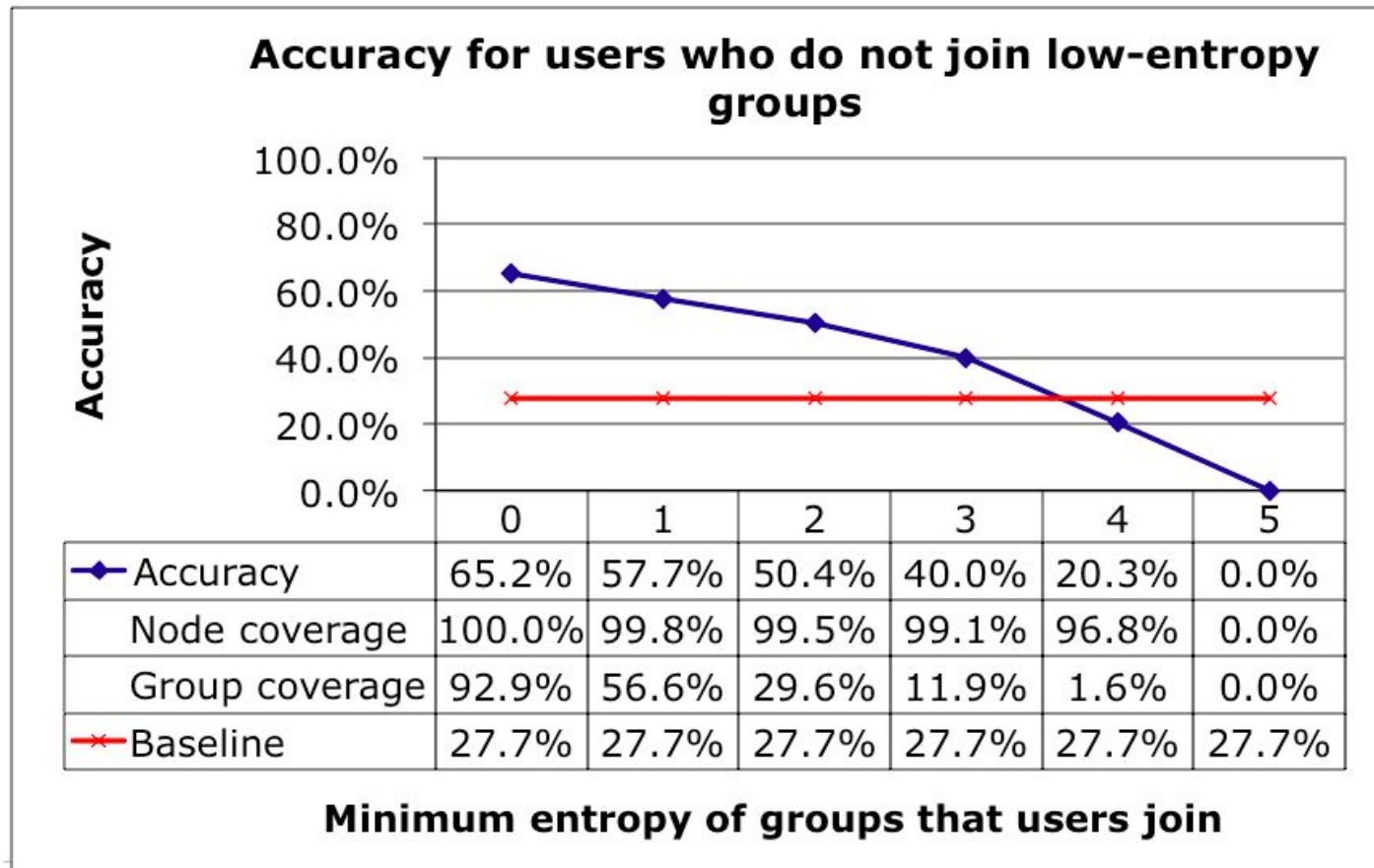
- Higher % public profiles -> higher accuracy





Your privacy depends on others

- Higher % public profiles -> higher accuracy





Main takeaways

- Increasingly challenging to manage personal information online
 - Probabilistic models can infer hidden information and circumvent privacy preferences of users
 - Groups are most significant carriers of private information
- For privacy-concerned users
 - If possible, set group and friendship lists as private
- For social media businesses
 - Enable greater user control over release of information
- For policy makers
 - Set standards for personal data release practices